Monodip Bhattacharya

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Summary

Full-stack (MERN) developer and B.Tech (ECE) graduate who builds real-time, secure, and well-tested web applications end to end. Strong in JavaScript/TypeScript, React, Node.js, and MongoDB, with hands-on experience in automated testing, CI/CD, and scalable real-time systems.

Education

Dr. BC Roy Engineering College, Durgapur — Durgapur, India
B.Tech in Electronics and Communication Engineering — *Aug 2022 – July 2026*
- CGPA: 6.63/10

Indira Gandhi Memorial High School — Kolkata, India
Class XII – CBSE — *March 2022*

St. Stephen's School — Kolkata, India
Class X – ICSE — *March 2020*

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3
Frameworks & Libraries: React.js, Node.js, Express.js, Socket.io, TailwindCSS, Zustand, Vite
Databases & Cloud: MongoDB, MySQL, Cloudinary, Render
Testing & DevOps: Jest, Supertest, GitHub Actions (CI/CD), Git, GitHub
Security, APIs & CS: REST APIs, JWT, bcrypt, Zod, Content Security Policy, Web Push, PWA, Data Structures & Algorithms

Internships

Embedded Systems & IoT Intern — June 2025 – July 2025
IC Design and Fabrication Center, Dept. of ECE, Jadavpur University — *Kolkata, India*
- Programmed Arduino microcontrollers to interface with sensors and transmit data over IoT protocols, building end-to-end hardware-software pipelines covering circuit design, firmware development, and IoT data flow architecture.
- Delivered a complete real-time IoT project within a 2-month timeline; received official certification upon project completion.

Projects

Donor | *React.js, Node.js, Express, MongoDB, Socket.io, TailwindCSS, Zustand, Jest, GitHub Actions* | GitHub | Live — 2026
- Built a real-time blood donation platform that matches compatible donors within 15 km of a hospital using MongoDB geospatial queries, with Socket.io and Web Push alerts, in-app chat, OpenStreetMap hospital search, and India's 120-day donation gap rule (G.S.R. 166(E)).
- Secured with JWT in HTTP-only cookies, bcrypt, Zod validation, rate limiting, and a strict Content Security Policy; built transactional account deletion and a DPDP Act-aligned privacy notice with versioned consent records.
- Shipped as an installable PWA with route-level code splitting and safe service-worker updates; 300+ Jest/Supertest tests run in a GitHub Actions CI pipeline that gates every deploy to Render.

AI Resume Analyzer | *React, Vite, PDF.js, Puter.js AI, REST APIs* | GitHub | Live — 2025
- Built an AI-powered resume analysis tool that parses uploaded PDFs with PDF.js and returns ATS scores, grammar feedback, keyword suggestions, and recommended job roles.
- Surfaced results in an analytics dashboard powered by Puter.js AI, giving users actionable, role-specific improvement guidance.

Additional

Interests: Football, Music, Reading, Travelling
Soft Skills: Problem-solving, Analytical Thinking, Time Management, Team Collaboration, Communication